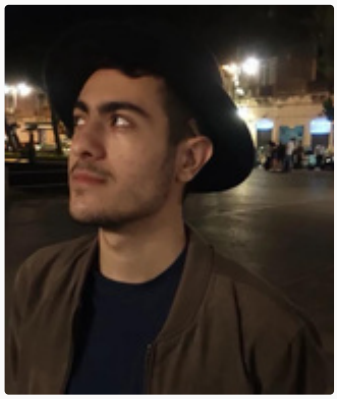


Claudio Giusti

Software Engineer

claudiogiusti07@gmail.com · https://claudiunderthehood.github.io

Professional Summary

Software Engineer with a strong background in hardware and software development. Experienced in aerospace hardware/software systems and user interface design as well as independent software development as a Freelancer. Passionate about computer science research, specializing in multimedia forensics and Artificial Intelligence with first-author Q1 journal publications. Currently pursuing a Master's degree in Computer Science.

Work Experience

FREELANCE SOFTWARE ENGINEER

Self-Employed | Italy | 2021 - Present

- Design and deliver custom software solutions and technical consulting for several customers.
- Develop software architectures, data processing pipelines, and code optimization strategies.
- Manage the full software development lifecycle independently, from requirements gathering to final deployment.

HARDWARE & SOFTWARE DEVELOPER

ItalSpazio | Catania, Italy | 2018 - 2021

- Developed hardware and firmware solutions for satellite devices and communication hardware.
- Designed and built intuitive user interfaces (UI) to streamline user interaction with satellite systems.
- Diagnosed and resolved complex hardware-software integration issues.

Education

M.Sc. in Computer Science | University of Catania (Oct 2026 - Present)

B.Sc. in Computer Science | University of Catania (Oct 2020 - July 2026)

High School Diploma in Electronic Engineering and Robotics | ITIS G. Marconi (2015 - 2020)

Languages

- Italian: Native
- English: Proficient / Native-level (Cambridge C2 Certification)
- Chinese: Elementary (HSK 2 Certification)
- Spanish: Beginner (A1 Level)

Scientific Publications

- **Proto-LeakNet: Towards signal-leak aware attribution in synthetic human face imagery**
  C. Giusti, L. Guarnera, S. Battiato - Computer Vision and Image Understanding (Elsevier), Vol. 270, 2026 (Q1 Journal)
- **Fraud is not just rarity: A causal prototype attention approach to realistic synthetic oversampling**
  C. Giusti, L. Guarnera, M. Casu, S. Battiato - Knowledge-Based Systems (Elsevier), Vol. 350, 2026 (Q1 Journal)

Core Skills

Software Engineering & Research

- Software Architecture & System Design
- Machine Learning & Computer Vision
- Deepfake Detection & Synthetic Media Forensics
- Explainable & Interpretable AI
- Generative AI

Technical Tools & Languages

- **Languages:** Python, C/C++, Java, C#
- **Frameworks:** Pytorch, Tensorflow, OpenCV
- **Tools:** GIT, Docker, Linux

Soft-Skills

- Research-Oriented Problem Solving
- Self-Management & Adaptability
- Collaborative Academic & Industrial Teamwork

Portfolio & Open-Source Projects

- **Proto-Leaknet (GitHub)**
- **VAEGAN-CPAC (Github)**
- **TetrisRLC (Github)**
- **ESP32DeautherTool (Github)**
- **TextMarkLab (Github)**
- **Github**

Interests

- **AI Research & Open-Source:** Tracking state-of-the-art advances in Deep Learning, Computer Vision, and Cybersecurity
- **Embedded Systems & Electronics:** Designing and prototyping custom micro-circuits and IoT hobby projects.

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